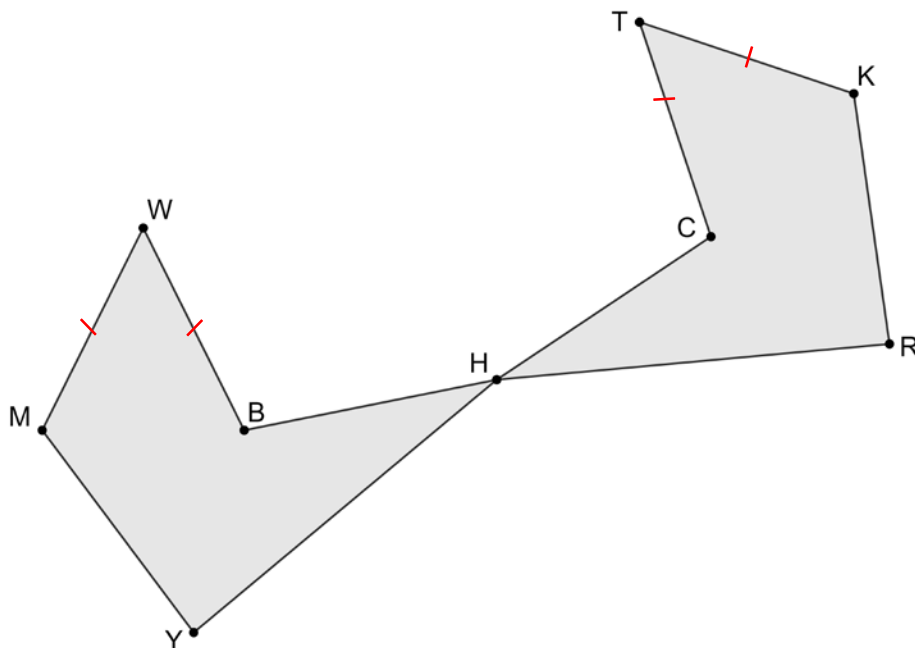


Congruent polygons $WBHYM$ and $TCHRK$ are shown where $\overline{TC} \cong \overline{TK}$.



- Determine the length of $\overline{TC}$ if $TC = 8x - 11$ and $TK = 3x + 24$.
 $8x - 11 = 3x + 24$
 $5x = 35$
 $x = 7$
 $TC = 8(7) - 11$
 $= 56 - 11$
 $= 45$
 $TC = 45$
- Determine the value of y when the perimeter of $WBHYM$ is 236 units and $HR = 7y - 6$, $KR = 5y - 11$, and $BH = 4y + 3$.
 $(7y - 6) + (5y - 11) + (4y + 3) + 45 + 45 = 236$
 $16y + 76 = 236$
 $16y = 160$
 $y = 10$
- What is the length of $\overline{HR}$?
 $HR = 7(10) - 6$
 $= 70 - 6$
 $= 64$
 $HR = 64$
- What is the length of $\overline{MY}$?
 $MY = KR = 5(10) - 11$
 $= 50 - 11$
 $= 39$
 $MY = 39$
- What is the length of $\overline{HC}$?
 $HC = BH = 4y + 3$
 $= 4(10) + 3$
 $= 43$
 $HC = 43$

Some of the angle measures of $WBHYM$ and $TCHRK$ are shown.

$$m\angle WMY = (8k - 19)^\circ$$

$$m\angle HRK = (5k + 2)^\circ$$

$$m\angle KTC = (3k + 2)^\circ$$

$$m\angle RHC = (2k - 6)^\circ$$

$$m\angle WBH = 9(m\angle RHC) + 3$$

$$m\angle RKT = (5k + 32)^\circ$$

6. Determine the value of k .

$$m\angle WMY = m\angle RKT$$

$$8k - 19 = 5k + 32$$

$$3k = 51$$

$$k = 17$$

Determine each angle measure.

7. $m\angle BWM = \underline{53^\circ}$

$$3(17) + 2$$

$$51 + 2 = 53$$

8. $m\angle WBH = \underline{255^\circ}$

$$9(2k - 6) + 3$$

$$18k - 54 + 3$$

$$18k - 51$$

$$18(17) - 51 = 255$$

9. $m\angle MYH = \underline{87^\circ}$

$$5(17) + 2$$

$$85 + 2$$

10. $m\angle TCH = \underline{255^\circ}$